Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	*(a)*	(i)	Total internal reflection Don't accept TIR	1			1		
		(ii)	$1.52 \sin \theta_C = 1.33$ [$\sin 90°$] (1) $\theta_C = 61°$ (1) 45° (or angle of inc) < 61°(or θ_C) so won't work properly **or** no TIR **ecf** (1) **Alternative:** $1.52 \sin 45° = 1.33 \sin \theta_w$ (1) $\theta_w = 54°$ or $\sin \theta_w < 1$ (1) Light escapes or not TIR so won't work properly **ecf** (1)			3	3	3	
	(b)	(i)	$t = \frac{d}{v}$ in this transposed form, or by implication – ignore units and accept omission of n (1) $v = \frac{c}{n}$ or equivalents used or by implication (1) 7.6 μ[s] (1)		3		3	3	
		(ii)	Monomode fibre allow only one route/path for light [whereas multimode fibres allow many] (1) [Therefore] any one pulse arrives without spreading [over time] (smearing) in a monomode fibre. Accept all light takes the same time to travel through fibre. Accept converse point for thicker (1) Hence likelihood of pulses overlapping [muddling] (1)	3			3		
			Question 6 total	**4**	**3**	**3**	**10**	**6**	**0**